Shi Tang

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EDUCATION

Imperial College London London, UK
MEng Electrical and Information Engineering *Sep 2025 – Present*

- Modules: Digital Electronics and Computer Architecture, C++ Programming for Engineers, Analysis and Design of Circuits, Engineering Mathematics.

EXPERIENCE

Jihua Laboratory Guangdong, China
Robotics Research Intern – Autonomous Mobile Robot Development *Aug 2026 - Sep 2026*

- Developed a ROS 2 Humble autonomous mobile-robot navigation system in Gazebo and deployed it on a WHEELTEC differential-drive chassis powered by an Intel NUC13, integrating LiDAR, an IMU, and wheel encoders.
- Implemented a LiDAR-based SLAM and Nav2 stack, fusing wheel odometry and IMU measurements through an EKF to support mapping, localisation, map saving/loading, and RViz 2D Goal Pose navigation.
- Validated the end-to-end system on the physical robot by mapping an office environment, saving and reloading occupancy maps, and performing autonomous navigation with local obstacle avoidance.

PROJECTS

Collision Avoidance and Rigid Body Segmentation
AI Training on Web Server *May 2025 – Jul 2025*

- Researched deep neural-network architectures for collision avoidance using stereo-RGB inputs from the KITTI autonomous-driving dataset.
- Implemented and trained the RigidMask model in PyTorch with U-Net and ResNet architectures for rigid-body segmentation; achieved 91.3% accuracy after model tuning.
- Generated segmentation outputs and predicted object-motion trajectories on a Linux-based web server, validating system behaviour and presenting results in a group setting.

Autonomous Driving Model Car
Embedded Vision and Control *Jul 2024 – Oct 2024*

- Designed and built a software-controlled autonomous vehicle model using a Raspberry Pi, Arduino and embedded C.
- Implemented real-time lane detection at 20 FPS using OpenCV to extract lane deviation from camera images.
- Developed control algorithms and interfaced with Arduino-based motor controllers for dynamic speed adjustment, enabling stable lane following on test tracks.

SKILLS

Technical Skills: ROS 2, Gazebo, RViz, OpenCV, PyTorch, YOLO, Python, SystemVerilog, C++, Linux, Git
Languages: Mandarin, English